

Phys 224 Problem Set 1

Due Thursday Nov. 10 at start of class. You may talk/consult with whom or whatever, but you must write your own solutions for submission.

1) Algebra practice: Solve $\sqrt{ab^2} = c + d$ for b .

2) Dimensional analysis: speed of something in air. Take the density and pressure of air. How can you combine them to make a speed? What is that speed? Compare it to the speed of sound.

3) a) Dimensional analysis 2: it may not come as a surprise that if you pluck a string under tension, it vibrates as displacements travel back and forth along the string. The two quantities that are relevant to a string are the tension T it is under (which has units of force), and the mass per unit length ρ of the string (which has units of kg/m). How can you combine T and ρ to make a speed?

b) String tension guides tell me that a violin E-string (the highest-pitched string) is under 18 or so pounds of tension, and a G-string (the lowest pitched one) is under about 10 pounds. What are these tensions in SI units?

c) As near as I can tell, the mass of the vibrating part of the E-string (the stretch between the neck and the bridge) is about 0.3 grams, and the vibrating mass of a G-string is around 2.5 grams. What are the approximate speeds you would expect to see signals moving up and down the two strings? Note that I have given you a mix of units, and there's one quantity I haven't given you that you'll have to estimate. Make sure to convert everything to a consistent set of SI units!

3) a) **Ideal gas practice.** On a cold winter's day, your professor cooks a large pot of warm, comforting soup. When the soup is done, your professor puts a lid on the gently boiling soup, and sticks it outside to cool down. He then forgets about it, and wakes up in the -10 C morning to find the soup is frozen solid. Assuming that the lid was on tightly (so no air could enter the pot once the lid is on), what is the pressure inside the pot in the morning?

b) What force would it take to lift up the lid in the morning to get at the cold-but-still-delicious soup? Feel free to estimate the size of the soup pot you think he deserves. (Note - this has actually happened to me, and the only way I could get the lid off was to bring the soup back to a boil.)

4) **OPTIONAL!** But maybe fun? I said in class that we aren't going to work through the details of how gasses heat up/cool when we compress/expand them. We can in fact work this out using the ideal gas law and the bit of statistical mechanics we already used.

a) In class, I claimed that the average energy of a particle moving only in the x - direction is $\frac{1}{2}k_B T$ where k_B is Boltzmann's constant (1.38×10^{-23} Joules/K) and T is the temperature of the gas. Of course, we live in 3-dimensional space, so the particle can move in the x -, y -, or z -directions. Each of these are independent, so the total "typical" kinetic energy of a molecule is $\frac{3}{2}k_B T$. Show

that the ratio of energy density in a gas to the pressure is $\frac{3}{2}$. What are the units (if any) of that ratio?

b) Let's say we have a box of air¹ with volume V , temperature T , pressure P etc. Then we squeeze the box so the volume goes down by some small amount, say 1% of the original volume. How much work did we do?

c) All of that work we did goes into heating the air in the box. What is the new temperature of the air inside the box in terms of the old temperature? Since the difference in energy is from the work we did, we know that

$$\frac{3}{2}k_B T_{new} = \frac{3}{2}k_B T_{old} + dW \quad (1)$$

where dW is the amount of work we did compressing the gas, and T_{old} and T_{new} are the initial and final temperature of the gas. Express the fraction by which T_{new} is different from T_{old} given the fraction we compressed the box.

d) Now that we know the new temperature, what is the new pressure in the box? If we kept the temperature fixed, if we compressed the box by 1%, then the pressure would go up by 1%. Because we heated the gas, you should find that the pressure goes up by more than 1%.

Discussion: In all the cases we'll talk about in this class, we'll find that indeed when we shrink the volume, the pressure goes up by more than we would have expected if we helped the temperature fixed. This ratio is usually constant, and is known as the *adiabatic index* (usually written γ) of a gas. The ideal gas law $PV = Nk_B T$ holds for both helium and air, but they have different adiabatic indices. This show up *e.g.* when we calculate the speed of sound in a gas. Because the correction factor is constant, we won't be *very* wrong if we ignore it, but if you want to actually make a detailed *correct* calculation, you should probably include the actual adiabatic index.

¹Or helium, if you're being persnickety. For actual air with molecules, the previous part would actually be 5/2 because the air molecules spin around. We'll stick with 3/2 for now.